

Mathematics AA vs AI

Mathematics: Analysis and Approaches (SL/HL)

This course recognizes the need for analytical expertise in a world where innovation is increasingly dependent on a deep understanding of mathematics. It places emphasis on constructing, communicating, and justifying correct mathematical arguments. Students who choose Maths AA at SL/HL level should be comfortable with using algebraic expressions, recognizing patterns, and understanding the mathematical generalization of these patterns.

Mathematics: Applications and Interpretation (SL/HL)

This course recognizes the increasing role that mathematics and technology play in a diverse range of fields in a data-rich world. Its extensive use of technology allows students to explore and construct mathematical models. Students who choose Maths AI at SL/HL level should enjoy using mathematics in real-world contexts and solving real-world problems.

Mathematics: Analysis and Approaches (SL/HL)

This course is mostly about theoretical maths. It contains abstract theories and covers topics like advanced calculus and integration to train students in the discipline of pure mathematics.

Mathematics: Applications and Interpretation (SL/HL)

Have you ever found yourself in a maths class, trying to understand complicated formulas, questioning if this can ever be used in real life? The course “Applications and Interpretations” is your answer. AI is all about the practical application of mathematical concepts.

Differences - Syllabus

Analysis and Approaches		Hours / Topic	Applications and Interpretation	
SL	HL		SL	HL
19	39	Number and Algebra	16	29
21	32	Functions	31	42
25	51	Geometry and Trigonometry	18	46
27	33	Statistics and Probability	36	52
28	55	Calculus	19	42

Differences - Assessment

Analysis and Approaches			Applications and Interpretation
SL	HL	SL	HL
No-calculator allowed. 80 marks, 90 mins. Weight – 40% Short/long questions.	No-calculator allowed. 110 marks, 120 mins. Weight – 30% Short/long questions.	Paper 1 Calculator allowed. 80 marks, 90 mins. Weight – 40% Short questions.	Calculator allowed. 110 marks, 120 mins. Weight – 30% Short questions.
Calculator allowed. 80 marks, 90 mins. Weight – 40% Short/long questions.	Calculator allowed. 110 marks, 120 mins. Weight – 30% Short/long questions.	Paper 2 Calculator allowed. 80 marks, 90 mins. Weight – 40% Long questions.	Calculator allowed. 110 marks, 120 mins. Weight – 30% Long questions.
	Calculator allowed 55 marks, 60 mins Weight – 20% 2 Long questions.	Paper 3	Calculator allowed. 55 marks, 60 mins. Weight – 20% 2 Long questions.
Mathematical exploration A piece of written work that involves investigating an area of mathematics (20 marks). Weight – 20%		IA	Mathematical exploration A piece of written work that involves investigating an area of mathematics (20 marks). Weight – 20%

How to choose the right course

There are 3 important things to consider:

- Your abilities and interests
- Your choice of other subjects in the Diploma Programme
- Your academic and career plans. The IB suggests:
 - **Maths AA** for students interested in mathematics, engineering, physical sciences, and some economics.
 - **Maths AI** for students interested in social sciences, natural sciences, medicine, statistics, business, engineering, some economics, psychology, and design.

Generally speaking, AA provides a **classical approach** to mathematics. If you are eager to dive into the world of abstract theories and generalizations, then AA is the right course for you!

On the other hand, if you like the idea of dealing with the practical applications of mathematics, then AI is a better fit.

Entry requirements

Final Grade 10 Semester 2 report grade out of 7:

To qualify for AA SL: 4, 5, 6, or 7 if in Ext. & 5, 6, or 7 if in St.

To qualify for AI SL: 4, 5, 6, or 7 if in Ext. & 4, 5, 6, or 7 if in St.

To qualify for AA HL: 6 or 7 if in Ext. & 7 if in St.

To qualify for AI HL: 6 or 7 if in Ext. & 7 if in St.